VAYU | YOUR PERSONAL CROP GUIDE

Dragon fruit (Kamalam)

How to grow it on your land

Prepared for: General guide (no location chosen)
Prepared on 2026-11-11

Hardy and low in pest problems, but needs poles/trellis up front and the market price varies a lot with fruit size and pulp colour.

DIFFICULTY	MONEY NEEDED	RISK	FIRST RETURN
3 of 5	**High**	**Medium**	**about 1-2 years from cuttings**

How to read this guide: [S1] marks a line backed by a source listed on the last page. (general knowledge) marks a line that has not been verified against a source. Lines starting **For your land** use the numbers for your location.

In this guide

- Your priorities: what to do first, in order
- Your planting plan: how much to buy and prepare for your plot
- How to grow it: planting, water, nutrition, pests and diseases, monitoring, harvest, risks
- If your conditions are different, and the sources behind every line

Your land at a glance

Your priorities

Ordered by what matters most on your land.

1. Get your soil tested
Satellite and model data estimate soil pH and texture only roughly, and cannot see nutrients, salinity or soil health. Get a proper soil test (pH, organic carbon, nutrients, salinity) from your KVK or the state soil-testing laboratory; the Soil Health Card scheme provides this. (general knowledge)

2. Talk to your KVK first
Take this guide to your nearest Krishi Vigyan Kendra and ask for their local advice and trusted planting material. (general knowledge)

3. Start small
Plant or build a small trial first. Learn the crop on your own land, then expand from what actually worked. (general knowledge)

4. Line up a buyer before you scale
Know who will buy, in what form and grade, and how it will reach them. (general knowledge)

Your planting plan for 1 acre

What	How much	Basis
Plants at 3 m x 3 m	**about 449**	your plot size x cited amounts [S2]
Posts with a ring on top (one per plant)	**449 posts**	your plot size x cited amounts [S2]
Farmyard manure or compost at planting	**4.5 to 6.7 tonnes**	your plot size x cited amounts [S1]

Sums of the cited per-plant and per-hectare amounts for your plot size. Real needs vary with soil and variety; confirm with your KVK.

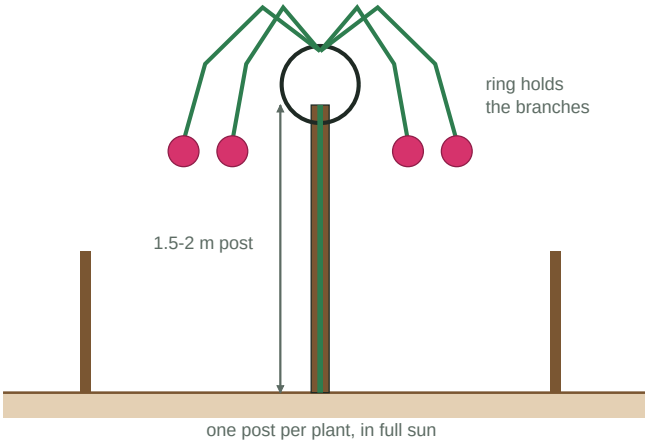
Planting material

- Propagate from stem cuttings of a healthy mother plant; ready-to-plant rooted cuttings take about 4-5 months in polybags (soil : FYM : sand). [S2]
- Buy only from a reputable nursery or an ICAR/KVK source; poor planting material spreads disease. (general knowledge)
- Expert guidance is available from ICAR-IIHR, ICAR-NIASM Baramati and ICAR-CIARI Port Blair. [S1, S3]

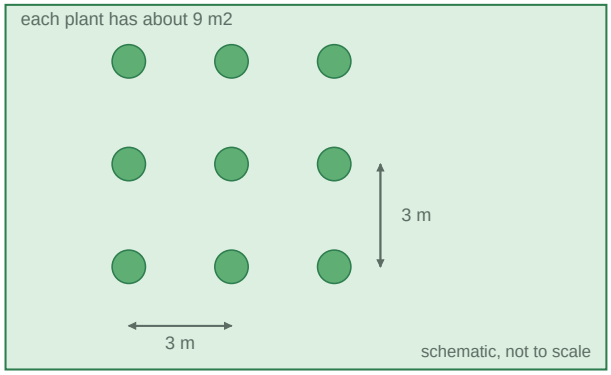
Land preparation and planting

- Full sun, well-drained soil. Sandy loam to loam rich in organic matter is best; all soil types can work if drained. [S2]
- Install a post (about 1.5-2 m high) with a ring on top for each planting spot; the plant climbs. Common spacing is 3 m x 3 m (single post); 2 m x 2 m is also used. [S2]
- Apply 10-15 kg farmyard manure per plant at planting. [S1]

- Keep water from standing around the base of the stem. [S2]



Post and ring support for a climbing cactus

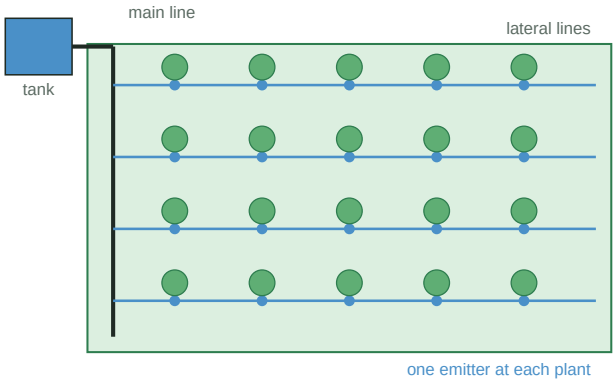


Plant spacing, top view

Irrigation

- Keep soil moist but never soggy; this is a cactus and excess moisture rots the stem. [S4]

- Drip irrigation with mulch saves water and keeps the base dry. (general knowledge)

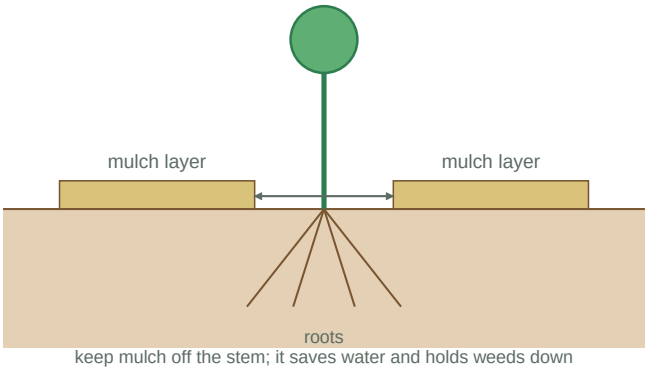


Drip irrigation layout (schematic)

Nutrition (organic)

- Rely on organic manure: farmyard manure, compost, neem cake; poultry manure can supplement. ICAR-IIHR notes it can be grown organically. [S1, S2]

- Re-apply compost or manure every few months and mulch around the plant. [S4]

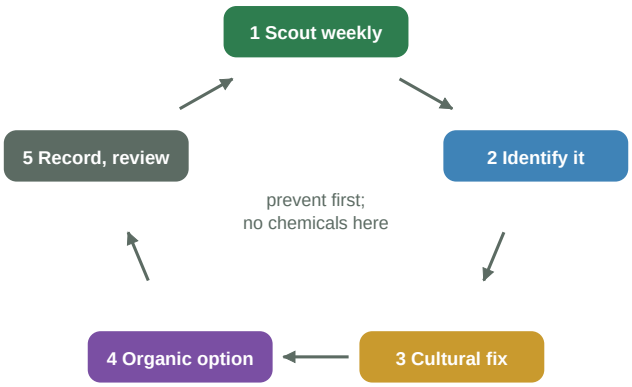


Mulch around a plant

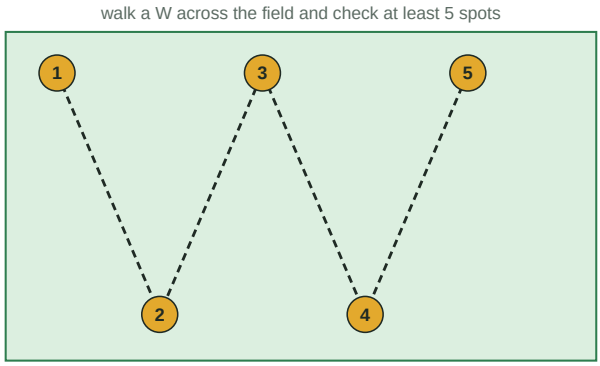
Pests and diseases (organic and cultural only)

- Stem rot and spots are the main problems and are linked to heavy rain and over-watering; prevent with drainage, clean tools and removing and destroying infected stems. [S1]
- Bag fruits to protect them from sunburn, birds and insects. [S1]
- A weed mat or mulch reduces weeds and conserves moisture. [S1]

- For anything beyond these organic and cultural steps, ask the nearest Krishi Vigyan Kendra (KVK) or state agricultural university for their current recommendation. (general knowledge)



Pest and disease loop: prevent first, then look, fix, review



How to walk the field when scouting

Monitoring

- Walk the field weekly: check stem bases for soft or discoloured patches, yellowing, sunburn marks, and new flower buds. (general knowledge)
- Note flowering flushes (typically in 3-5 staggered flushes, June to November) and harvest 30-35 days after flowering. [S5]

Harvest, yield and post-harvest

- Fruits are ready about 30-35 days after flowering; weight is typically 200-700 g depending on management. [S5]
- Harvest season in India is mainly June to October. [S4]
- Yellow varieties can be stored 28-30 days at 10 C. [S6]

Risks and costs

- Initial cost is mainly poles/trellis and planting material, so capital is needed before the first fruit. (general knowledge)
- Market price is highly variable and depends on fruit size and pulp colour. [S5]
- India's area is growing, so check local demand and a buyer before planting a large area. (general knowledge)

If your conditions are different

What to do if your land, or the season, turns out different from the numbers above.

- **Heavy soil** (fixable): Sandy loam to loam is best, and any soil can work if it drains. Heavy clay holds water around the stem and invites rot, so plant on raised beds or mounds and mix in organic matter. [S2]
- **Cold weather** (fixable): Freezing injures the plant; it is not suitable for temperate or frost-prone areas. Where winters are only mildly cool, protect young plants; do not plant where hard frost is normal. [S2]
- **Heat** (fixable): Very high temperature for long periods (above about 40 C) harms growth, flowering and fruit. Use a shade net over young plants and fruit, and mulch the base. [S2]
- **Low rainfall** (fixable): Low rainfall is workable (commercial crops reported from about 400-500 mm) with drip irrigation and mulch. [S2, S3]

- **Too much rain** (fixable): Heavy rain and waterlogging cause stem rot. Plant on raised beds or mounds with drainage; avoid low-lying land. [S1, S2]
- **Soil pH** (fixable): Bring pH toward 5.5-7.0 with compost and farmyard manure; get a soil test before adding any amendment. (general knowledge)

About chemicals

This guide gives organic and cultural methods only. It does not name or dose pesticides, fungicides or chemical fertilisers; for those, follow your nearest Krishi Vigyan Kendra (KVK) or state agricultural university and the label of any registered product.

Sources

[S1] https://en.vikaspedia.in/agriculture/crop-production/package-of-practices/fruits-1/dragon-fruit
[S2] https://epubs.icar.org.in/index.php/IndFarm/article/download/174026/63862/503179
[S3] https://epubs.icar.org.in/index.php/IndHort/article/download/107290/42259/277997
[S4] https://vikaspedia.in/agriculture/crop-production/package-of-practices/fruits-1/dragon-fruit
[S5] https://www.downtoearth.org.in/agriculture/dragon-fruit-is-it-the-next-big-thing-in-indian-horticulture-78017
[S6] https://en.vikaspedia.in/viewcontent/agriculture/crop-production/package-of-practices/fruits-1/dragon-fruit?lgn=en

Advisory aid from published sources and general agronomic knowledge, not a guarantee of yield or income and not a substitute for local extension advice. Lines marked 'general knowledge' are not verified against a source.